Mani Saideep Chintha

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github.com/mani20030606 | linkedin.com/in/mani-saideep-chintha | Portfolio

Professional Summary

AI/ML and Computer Vision Engineer with hands-on experience training, evaluating and deploying object detection, segmentation and tracking models (YOLO, SAM, ByteTrack) using PyTorch, TensorFlow and OpenCV, plus RAG/LLM-based GenAI pipelines. Skilled in dataset annotation, model evaluation (mAP, precision/recall) and edge and API deployment (ONNX, TFLite).

Experience

Computer Vision Engineering Intern – Huebits, Visakhapatnam *Jan 2026 – Present*

- Promoted from a campus-level internship to a paid Computer Vision Engineering role based on performance.
- Lead and contribute to production computer vision projects, including dataset preprocessing, model training, evaluation, and deployment.
- AI granite measurement app won 1st prize, a cash prize and a work order.

Machine Learning Intern – Skill Development Program, Andhra Pradesh

- Built regression, classification, and clustering models in Python, achieving 85%+ accuracy across multiple real-world datasets through feature engineering and hyperparameter tuning.

Projects

AI Granite Block Measurement & Seigniorage Platform **HUEBITS**

Project Lead (as CV Engineering Intern) | Python, YOLO, SAM, OpenCV ArUco Marker, ONNX, TFLite

- **Won 1st Prize at the AP RTGS AI Hackathon** and received a work order with a cash prize from the Andhra Pradesh Government Mining Department.
- Led the CV pipeline (YOLO segmentation, SAM mask refinement, ArUco homography) that replaces manual tape measurement with photo/video-based volume estimation at **95% accuracy** against ground-truth measurements.
- Trained on 1,000+ self-collected and labeled images; exported to ONNX and TFLite for offline Android inference; built a 4-face capture workflow with cross-checks to prevent fraud.

Highway Accident Prevention **HUEBITS** | *Ongoing*

Python, YOLO, OpenCV

- Developing computer vision models using YOLO for vehicle detection, pothole detection, and signage board detection.

AI-Based Firearm Detection from Surveillance Footage **Personal project**

Python, YOLOv8, ByteTrack, Deep Learning, Flask

- Trained YOLOv8 on a 16,000+ image annotated and augmented dataset, raising accuracy from 78% to 91% mAP@0.5 and cutting false positives by 20% through precision/recall-based threshold tuning.
- Integrated ByteTrack for persistent object IDs across frames and deployed a Flask REST API for inference at under 300ms latency and 20+ FPS.

Hybrid GraphRAG Intelligence System **Personal project**

Python, Neo4j, ChromaDB, LLMs

- Built a hybrid GraphRAG system (Neo4j knowledge graphs + ChromaDB vector search) with RRF fusion and a 3-tier LLM fallback for low-hallucination, cited Q&A.

Skills

- **Languages:** Python, JavaScript, HTML, CSS
- **Computer Vision:** OpenCV, YOLOv8/v11, Object Detection, Multi-Object Tracking (ByteTrack), OCR (Tesseract), Image Segmentation (SAM), Camera Calibration, ArUco Pose Estimation, ONNX, TFLite
- **Data:** Dataset Annotation & Labeling, Data Augmentation, Exploratory Data Analysis, Model Evaluation
- **AI/ML:** PyTorch, TensorFlow, Keras, Transfer Learning, Regression, Classification, Clustering, Model Monitoring
- **LLM/GenAI:** RAG Pipelines, LLM Orchestration, Prompt Engineering, Hugging Face Transformers, Ollama
- **Tools & Infrastructure:** FastAPI, Flask, Neo4j, ChromaDB, PostgreSQL, MySQL, MongoDB, Docker, Git

Education

B.Tech – Computer Science (CSM) 2023 – 2026

Sasi Institute of Technology and Engineering, Tadepalligudem CGPA: 8.15

Certifications

- OpenCV BOOTCAMP – OpenCV| OCI AI Foundations Associate – Oracle | Python – SkillDzire | Machine Learning – SkillDzire

Honors & Awards

- **1st Prize, AP RTGS AI Hackathon (AP Govt. Mining Dept.)** Led development of an AI granite block measurement app that won **1st Prize at the AP RTGS AI Hackathon** and secured a **work order with a cash prize from the Andhra Pradesh Government Mining Department**
- **Promoted to Paid Intern** – Advanced to a paid, onsite CV role at Huebits after a campus-level round; contributed to a government-affiliated AI project.